

Published in final edited form as:

Int J Epidemiol. 2008 April ; 37(2): 301–308. doi:10.1093/ije/dyn002.

Maternal iron intake and iron status during pregnancy and child blood pressure at age 3 years

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Abstract

Background—Animal data suggest that maternal iron deficiency during pregnancy leads to lower birth weight and sustained blood pressure elevation in the offspring. In humans, iron deficiency during pregnancy is common and is associated with adverse birth outcomes such as low birth weight. Data are lacking, however, regarding the effects of maternal iron intake and iron status during pregnancy on offspring blood pressure. Our aim was to examine the extent to which lower maternal iron intake, haemoglobin level and mean cell volume (MCV) during pregnancy are associated with higher child systolic blood pressure (SBP) at age 3 years.

Methods—We studied 1167 participants in Project Viva, a longitudinal cohort study of pregnant women and their children. We estimated first and second trimester maternal iron intake from food frequency questionnaires. We used an electronic laboratory database to identify haemoglobin and MCV levels in pregnancy. We measured child BP up to five times with a Dinamap and used mixed-effects regression models in our analysis.

Results—Mean (SD) child SBP at 3 years was 92.0 (9.9) mmHg. Adjusting for confounders, for each 10 mg increase in first trimester iron intake, child SBP was not lower, but was in fact 0.4 mmHg higher (95% CI 0.1, 0.7). For second trimester iron intake, and for first or second trimester haemoglobin and MCV levels, we did not find any appreciable association with 3 year SBP.

Conclusions—In contrast to animal studies, we did not find that lower maternal iron status during pregnancy was associated with higher offspring BP.

Keywords

Blood pressure; hypertension; maternal nutrition; iron; foetal programming

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Introduction

Barker has proposed that poor foetal nutrition may contribute to the development of hypertension through foetal programming.¹ While numerous studies have focused on birth weight as a proxy for foetal nutrition, there are relatively few data regarding the influence of specific prenatal nutritional factors on offspring blood pressure.² Identifying such factors might inform public health strategies to prevent hypertension.

In rat models of blood pressure programming, maternal iron restriction during pregnancy leads to lower-weight offspring with sustained blood pressure elevation.³⁻⁵ In humans, maternal iron deficiency, anaemia and low iron intake are common during pregnancy^{6,7} and are associated with lower offspring birth weight due to foetal growth restriction^{8,9} and to premature birth,^{10,11} though some studies have not confirmed these associations.¹²⁻¹⁴ Modest iron supplementation may improve foetal growth and lengthen gestation.^{15,16}

Few studies in humans have examined the association between measures of maternal iron status during pregnancy and offspring blood pressure, and the existing data are inconsistent. Results from two studies examining maternal haemoglobin level during pregnancy in relation to child blood pressure suggest that child blood pressure is higher in children born to mothers with lower haemoglobin levels or who are anaemic, though one of these studies¹⁷ was small ($n = 77$) and in the other,¹⁸ the 95% confidence interval included both negative and positive associations. In contrast to these findings of an association of lower maternal haemoglobin or anaemia with higher offspring blood pressure, Whincup *et al.*¹⁹ examined maternal haemoglobin level and change in mean cell volume (MCV) during pregnancy and found no association with child blood pressure at 9–11 years of age; and Bergel *et al.*²⁰ found that child systolic blood pressure at 5–9 years was 1.3 mmHg higher for each g/dl greater maternal haemoglobin level during pregnancy. We are not aware of a study that has examined maternal iron intake during pregnancy in relation to child blood pressure. Thus, while maternal iron intake and iron status may plausibly influence offspring blood pressure in humans, and some existing evidence is suggestive of an association, further data from prospective studies are needed before recommendations for clinical practice can be considered.

The objective of this study was to examine the extent to which maternal iron intake and haemoglobin and MCV levels during pregnancy are associated with offspring blood pressure at age 3 years. We examined associations separately for the first and second trimesters and for iron intake from foods and supplements.

Methods

Study design and participants

We studied participants in Project Viva, a prospective, longitudinal cohort study designed to examine multiple prenatal factors in relation to outcomes of pregnancy and child health. We recruited participants into Project Viva at their first prenatal visit at Harvard Vanguard Medical Associates, a large multi-specialty group practice in eastern Massachusetts. Exclusion criteria included multiple gestation, inability to answer questions in English, plans to move from the area before delivery, and gestational age greater than 22 completed weeks at the initial prenatal clinical appointment. Additional details of recruitment and follow-up have been published elsewhere.²¹

Of the 2128 children in the original cohort, 549 were not eligible for the 3-year examination, either due to missing first and second trimester dietary data, or because the mother did not consent for the child to be enrolled in the study prior to the 3-year study visit. Of the 1579 children eligible for the 3-year study visit, at the time of this analysis 182 had not yet reached

the eligible age and mothers of 28 had not consented. Of the remaining 1369, we were unable to measure child blood pressure in 195, primarily due to parent or child refusal or because the parent completed a mailed questionnaire but the child did not attend an in-person study visit. We excluded an additional six children with implausible blood pressure measurements, and one child whose mother had did not have iron intake or haematological data, leaving a total sample of 1167 mother-child pairs.

For the iron intake analyses, we included the 1098 children whose mothers had completed the food frequency questionnaire administered in the first trimester, and the 1034 children whose mothers had completed the second trimester food frequency questionnaire. For the haemoglobin and MCV analyses, we included the 1047 children whose mothers had screening haemoglobin and MCV levels before 15 completed weeks of pregnancy (first trimester), the 996 with screening haemoglobin levels between 15 and 28 completed weeks (second trimester), and the 994 with screening MCV levels between 15 and 28 completed weeks. Human subjects committees of Harvard Pilgrim Health Care, Brigham and Women's Hospital and Beth Israel Deaconess Medical Center approved the study protocols.

Measurements

Maternal iron intake—We estimated maternal daily iron intake from diet and supplements separately, and calculated total daily iron intake (iron from diet plus supplements). To estimate maternal iron intake from diet, at study visits at the end of the first and second trimesters we administered a semi-quantitative food frequency questionnaire validated for use in large cohort studies including the Nurses' Health Study,^{22,23} and slightly modified for use during pregnancy.²⁴ In the first trimester, we asked participants to report intake 'during this pregnancy'; in the second trimester, we asked participants to report intake 'in the past 3 months'. At the end of the first trimester, we administered an additional interview to assess periconceptual and first trimester supplement use, including prenatal vitamins and iron supplements. Second trimester supplement use was included in the food frequency questionnaire. We estimated iron intake from supplements in mg elemental iron, the amount of iron in a supplement that is available for absorption. We assessed maternal diet and supplement use at two separate time points during pregnancy because nutritional exposures may have different effects on the offspring depending on when they occur. We used the Harvard Nutrient Database to estimate iron and other nutrient intake from foods.²² We considered low iron intake to be <27 mg of elemental iron per day, the Recommended Dietary Allowance during pregnancy.²⁵

Maternal iron status—Our two measures of maternal iron status were haemoglobin level and MCV, both of which prenatal providers measure routinely during pregnancy to screen for anaemia. A decrease in haemoglobin level occurs in more extreme cases of iron deficiency. We also examined MCV, which may decline despite a preserved haemoglobin level in milder cases of iron deficiency, as a secondary measure of maternal iron status. We identified laboratory results for each participant using the Harvard Vanguard Medical Associates clinical laboratory electronic database. Clinical laboratories used a Coulter Model S impedance-based haematology analyzer to measure haemoglobin and MCV levels. Some participants had multiple measurements of haemoglobin and MCV. To minimize possible selection bias, we identified for inclusion in our analysis screening laboratories, rather than follow-up laboratory studies, by using the haemoglobin and MCV that were ordered and/or reported on the same date as other screening laboratory studies in the first trimester (rubella serology, rapid plasma reagin, hepatitis B surface antigen) and second trimester (glucose challenge test). Our first trimester analyses included participants' screening laboratory test results obtained through 15 completed weeks gestation; second trimester analyses included results from greater than 15 through 28 completed weeks gestation.

We used the Centers for Disease Control and Prevention (CDC) criteria to define anaemia during pregnancy.²⁶ These cutoffs are haemoglobin <11 g/dl in the first trimester and haemoglobin <10.5 g/dl in the second trimester. The cutoff in the second trimester is lower than in the first trimester due to physiological expansion of maternal plasma volume during pregnancy. We could not identify pregnancy-specific norms for MCV, so we used the standard adult cutoff of 85 femtolitres (fl).^{26,27} Physiological expansion of maternal plasma volume is not likely to affect the MCV, thus we used the same cutoff throughout pregnancy.

Blood pressure—Research assistants measured child blood pressure at age 3 years with a Dinamap (Critikon, Inc) Pro100 automated oscillometric recorder by taking up to five measurements 1 min apart in each child, recording the child's state (sleeping, quiet awake, crying, active awake), position (sitting, semi-reclining, reclining, standing), extremity used, cuff size and room temperature.

Covariates—We examined covariates that could influence maternal iron status, child blood pressure or both. We collected data regarding maternal demographics and social, economic and health status through self-administered questionnaires and interviews by project staff at study visits during pregnancy, shortly after delivery, and at age 6 months and 3 years of age. Details about the source of information for these variables have been described elsewhere.^{21,28} We defined hypertensive disorders during pregnancy according to published standards.²⁹ Project staff used research standard methods to measure the mother's blood pressure and both the mother's and child's weight and height at the 3-year study visit.

Analysis

Our main outcome of interest was systolic blood pressure at 3 years of age. We used systolic blood pressure in the analysis because it predicts later outcomes better than diastolic blood pressure³⁰ and is measured more accurately with the Dinamap.³¹ To examine multivariable associations between maternal iron status and child systolic blood pressure, we used mixed effect regression models,³² incorporating each blood pressure measurement as a repeated measure. The advantage of this method is that children with fewer measurements and more variability contribute less weight in the analysis. In all models, we adjusted for blood pressure measurement conditions (child state and position, extremity used, cuff size) to minimize measurement error, as well as for child age and sex. In the iron intake analyses, we controlled for possible confounding by energy intake by the nutrient residuals method.²³ In the haemoglobin and MCV analyses, we adjusted for gestational age at the time of blood draw as a proxy for the expansion of plasma volume that occurs during pregnancy.

In our multivariable models, we adjusted for maternal age, sociodemographic variables (race/ethnicity, income, education), smoking status, and variables related to maternal nutritional status before and during pregnancy (pre-pregnancy body mass index and pregnancy weight gain), as well as child weight and height at age 3 years. We performed data analyses using SAS version 9.1 (SAS Institute Inc., Cary, NC, USA).

Results

Table 1 lists characteristics of participating mothers and children. Mothers of 27% of participants identified themselves as being from a racial or ethnic minority. Most mothers had at least a college education and an annual household income of greater than \$70 000. The large majority (94%) of children were born at greater than or equal to 37 completed weeks gestation and the mean (SD) gestational age was 39.5 (1.7) weeks. The mean (SD) birth weight was 3.50 (0.6) kg. At age 3 years, mean (SD) systolic blood pressure was 92.0 (9.9) mmHg and mean

(SD) diastolic blood pressure was 58.1 (7.8 mmHg), which are consistent with results of another study of blood pressure at this age measured with a Dinamap.³³

Compared with the original cohort of 2128 live births, mothers included in this analysis had similar mean pre-pregnancy body mass index and pregnancy weight gain, but were better educated (72 vs 65% with at least a college degree), had higher income (61 vs 54% with annual household income $\geq$ \$70 000), were less likely to be of a racial or ethnic minority (27 vs 33% non-white), and were slightly less likely to have smoked within 3 months before pregnancy (10.2 vs 11.7%). Compared with the entire cohort, children included in this analysis had a similar mean birth weight (3.50 vs 3.46 kg) and gestational age (39.5 vs 39.4 weeks) but fewer were born at less than 37 completed weeks gestation (5.9 vs 7.2%). Compared with the entire cohort, mothers included in this analysis had similar mean second trimester daily iron intake (50.4 vs 50.6 mg) and mean first trimester (12.7 vs 12.6 g/dl) and second trimester (11.6 vs 11.5 g/dl) haemoglobin levels; and slightly higher mean first trimester daily iron intake (34.4 vs 33.8 mg).

Measures of maternal iron status are detailed in Table 2. In the first trimester, mean (SD) total daily iron intake was 34.4 (17.2) mg and 33.2% of women had low iron intake. In the second trimester, mean total daily iron intake was 50.4 (25.2) mg, and 10.8% of women had low iron intake. Iron intake from supplements nearly doubled between the first and second trimesters, increasing from a mean (SD) of 16.8 (14.9) mg/day to a mean of 32.3 (23.5) mg/day. This increase likely reflects a shift in intake from multivitamins to prenatal vitamins and an increase in the proportion of women taking additional iron supplements from 5.0 to 14.9%. In the first trimester, mean (SD) maternal haemoglobin concentration was 12.7 (0.9) g/dl, mean (SD) MCV was 89.4 (4.8) fl, 2.7% of women were anaemic and 11.7% of women had a low MCV. In the second trimester, mean (SD) haemoglobin concentration was 11.6 (0.8) g/dl and 8.6% of women were anaemic. Mean (SD) MCV was 91.3 (5.1) fl and 8.2% of the women had a low MCV.

Table 3 describes results from our multivariable analyses. Adjusting only for blood pressure measurement conditions (extremity, cuff size, child's state, position and measurement order) and child age and sex, we found that child systolic blood pressure was 0.4 mmHg (95% confidence interval 0.1, 0.7) higher for each 10 mg increment increase in total maternal elemental iron intake during the first trimester. Additional adjustment for maternal age, pre-pregnancy body mass index, pregnancy weight gain, smoking status, income, education and race/ethnicity and child height and weight did not change this estimate appreciably. Child systolic blood pressure was 0.4 mmHg (95% confidence interval 0.1, 0.8) higher for each 10 mg increment increase in first trimester maternal iron intake from supplements only, similar to the estimate for total iron intake. We noted small, positive associations of second trimester iron intake from food and first trimester haemoglobin level with child systolic blood pressure; and a small, inverse association of second trimester MCV with child systolic blood pressure. Confidence limits for these three associations did not exclude a null effect. We found no association of first trimester iron intake from food, second trimester iron intake (total or from supplements), second trimester haemoglobin or first trimester MCV with child systolic blood pressure.

In secondary analyses, we examined differences in the systolic blood pressure of children born to mothers with and without anaemia, low MCV and low iron intake (Table 4). After adjustment for the same variables as in our primary analyses, we generally found only small differences in child systolic blood pressure. Children born to mothers without anaemia in the first trimester had 1.6 mmHg lower systolic blood pressure than children born to nonanaemic mothers, but the 95% confidence interval was wide (−5.0, 1.8). To allow comparison with other studies,^{17,18,20} we repeated our main analyses using as exposure variables the lowest maternal

haemoglobin and MCV levels and the change in MCV level from the first to second trimester, rather than the screening levels, and found similar results to our primary analyses (data not shown).

Discussion

Animal studies of severe maternal iron restriction during pregnancy support the hypothesis that low maternal iron intake or anaemia during pregnancy leads to higher blood pressure in the offspring through the biological process of developmental programming.^{3,4} Contrary to the relevant animal data and to our own study hypothesis, we did not find any inverse relationships between numerous measures of maternal iron status during pregnancy and child blood pressure at age 3 years.

Our findings are in agreement with a study in the United Kingdom¹⁹ that found no relationship between the minimum maternal haemoglobin level or change in MCV during pregnancy and child blood pressure at age 9–11 years. Similarly, an Argentinian study²⁰ found that child blood pressure was 1.3 mmHg higher for each 1 g/dl higher maternal haemoglobin level (95% confidence interval 0.4, 2.3) with a 95% confidence interval that excludes an inverse association between maternal haemoglobin level and child blood pressure and includes our effect estimate.

In contrast to our findings, Godfrey *et al.*¹⁷ found in a small ($n = 77$) Jamaican cohort that child systolic blood pressure at age 10–12 years was 2.6 mmHg higher for each 1 g/dl lower maternal haemoglobin level during pregnancy. In another UK study,¹⁸ systolic blood pressure was 2.8 mmHg higher in 4-year old children whose mothers' lowest haemoglobin level during pregnancy was <10 g/dl, compared with children whose mothers' lowest haemoglobin level was >12 g/dl, though the 95% confidence interval around the effect estimate (−3.4 to 1.7) did not exclude a null or even positive relationship between maternal haemoglobin and child blood pressure.

One possible reason for our finding of no relationship between maternal iron status and child blood pressure is that more extreme iron restriction and/or anaemia than was present in our cohort may be required for measurable effects to occur. In the relevant animal models, the degree of maternal iron restriction and anaemia induced in the experimental animals were more extreme than what we observed in our cohort. Both the Godfrey and the Law study cohorts included a higher proportion of women with anaemia than in our study, and may better represent the lower end of the spectrum of anaemia during pregnancy. Our study included very few women with a haemoglobin level less than 10 g/dl, the cutoff used in both of these studies.

Animal studies suggest that the programming effects of maternal iron restriction and iron deficiency on offspring blood pressure may not be evident until around puberty.^{3,5} In this study, we measured blood pressure in pre-school age children. Thus, we cannot exclude the possibility that an association of maternal iron intake or iron status with offspring blood pressure will become evident as the children become older.

Our study was limited by the high socioeconomic status and limited racial and ethnic diversity of our participants, which may limit the generalizability of our findings. Additionally, due to loss to follow-up, we studied only a subset of the original Project Viva cohort. Compared with the original cohort, fewer participants in our analysis were from racial or ethnic minorities or from lower socioeconomic strata, so selection bias is theoretically possible.

Other potential limitations are our measures of iron intake and iron status. Given the resources that would have been required, we were unable to assess iron intake by repeated food records or 24 hour recall. Assessing iron intake by food frequency questionnaire provides a reasonable

compromise between accuracy and feasibility for relatively large epidemiologic studies such as ours. In the absence of inflammation, the serum ferritin concentration provides the best non-invasive indicator of iron deficiency, but it is not measured routinely during pregnancy. The haemoglobin level and MCV are reasonable surrogates marker of iron status,²⁶ with complementary strengths and weaknesses. The haemoglobin level falls only in the later stages of iron deficiency, while the MCV decreases in earlier stages of iron deficiency. A low MCV, however, may be due to conditions other than iron deficiency. Additionally, one must consider the physiological decline in haemoglobin level that occurs in the first and second trimesters due to the expansion of maternal blood volume, which we did by using cutpoints for anaemia that differ by trimester and by adjusting for gestational age at the time of blood sampling. We were not able to assess iron intake or iron status during the third trimester, but animal data suggests that early iron supplementation in anaemic mothers in the early stages of pregnancy is more effective in improving foetal growth than supplementation at the end of pregnancy.³⁴

Ours is the largest study of the association between maternal iron status and child blood pressure, and the only study to measure maternal iron intake prospectively during two trimesters of pregnancy. We also performed careful measurement of our outcome, child systolic blood pressure at age 3 years, as well as numerous relevant covariates.

In conclusion, we did not find an association between maternal iron status during pregnancy and child blood pressure. While there are several reasons why pregnant women need adequate iron intake, reducing childhood blood pressure does not appear to be one of them, at least in well-nourished populations.

Acknowledgements

Funding was provided by NIH (HD 34568, HL 64925, HL 68041), Harvard Medical School, Harvard Pilgrim Health Care Foundation, Harvard Pediatric Health Services Research Fellowship Program (HRSA T32 HP10018).

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Table 1
 Characteristics of 1167 mother-child pairs in Project Viva

Mothers	Mean (SD)
Age (years)	32.5 (5.0)
Pre-pregnancy BMI (kg/m ²)	24.6 (5.2)
Pregnancy weight gain (kg)	15.7 (5.4)
SBP 6 months post partum (mmHg)	109.5 (10.9)
DBP 6 months post partum (mmHg)	64.8 (7.8)
Parity	Number (%)
0	547 (46.9)
1	419 (35.9)
> or = 2	201 (17.2)
Education	
> or = high school	82 (7.0)
Some college	245 (21.1)
College graduate	437 (37.5)
Graduate degree	400 (34.4)
Income	
<\$40 000	135 (11.6)
\$40 001–70 000	246 (21.1)
>\$70 000	716 (61.4)
Smoked within 3 months of learning of pregnancy	115 (10.2)
Race/ethnicity	
Black	135 (11.6)
Hispanic	70 (6.0)
Asian/other/more than 1	110 (9.5)
White	849 (72.9)
Children	Mean (SD)
Birth weight (kg)	3.50 (0.6)
Gestational age (weeks)	39.5 (1.7)
Weight at 3 years (kg)	15.8 (2.5)
Height at 3 years (cm)	97.4 (4.5)
SBP at 3 years (mmHg)	92.0 (9.9)
DBP at 3 years (mmHg)	58.1 (7.8)
	Number (%)
Male	592 (50.7)
Preterm (born at <37 completed weeks gestation)	69 (5.9)
Small for gestational age (birth weight for gestational age <10 th %ile)	60 (5.2)

Table 3

Multivariable mixed effects models showing the increment in 3-year systolic blood pressure (mmHg) per increment change in marker of maternal iron status

	1st Trimester			2nd Trimester		
	Change in SBP (mmHg)	95% CI	P-value	Change in SBP (mmHg)	95% CI	P-value
Iron intake, effect per 10 mg						
Total iron intake						
Crude	0.4	0.1, 0.7	0.02	0.0	-0.2, 0.2	0.91
Multivariable	0.4	0.1, 0.7	0.01	0.0	-0.2, 0.2	0.90
From food only						
Crude	-0.1	-1.0, 0.9	0.91	0.2	-0.8, 1.1	0.74
Multivariable	0.2	-0.8, 1.2	0.71	0.3	-0.7, 1.3	0.52
From supplements only						
Crude	0.4	0.1, 0.7	0.01	0.0	-0.2, 0.2	0.97
Multivariable	0.4	0.1, 0.8	0.01	-0.0	-0.2, 0.2	0.99
Haemoglobin, effect per g/dl						
Crude	0.4	-0.3, 1.0	0.20	0.3	-0.3, 1.0	0.28
Multivariable	0.5	-0.2, 1.1	0.17	0.2	-0.5, 0.9	0.47
MCV, effect per 10 fl						
Crude	0.0	-1.1, 1.1	0.97	0.0	-1.0, 1.1	0.95
Multivariable	-0.2	-1.4, 1.0	0.76	-0.3	-1.4, 0.9	0.65

CI is confidence interval. MCV is mean cell volume. Iron from supplements is in mg of elemental iron. For crude iron intake analyses, $n = 1095$ in the 1st trimester and $n = 1031$ in the 2nd trimester. For multivariable iron intake analyses, $n = 1044$ in the 1st trimester and $n = 983$ in the 2nd trimester. For crude haemoglobin analyses, $n = 1044$ in the 1st trimester and $n = 993$ in the 2nd trimester. For multivariable haemoglobin analyses, $n = 982$ in the 1st trimester and $n = 944$ in the 2nd trimester. For crude MCV analyses, $n = 1044$ in the 1st trimester and $n = 991$ in the 2nd trimester. For multivariable MCV analyses, $n = 982$ in the 1st trimester and $n = 942$ in the 2nd trimester. All analyses are adjusted for blood pressure measurement conditions (extremity used, cuff size, state, position, measurement order), child sex and age at measurement. Multivariable analyses are additionally adjusted for maternal age, pre-pregnancy body mass index, pregnancy weight gain, smoking, income, education and race/ethnicity, and child height and weight at the time of blood pressure measurement. Iron intake analyses are also adjusted for total energy intake; and haemoglobin and MCV analyses are adjusted for gestational age at blood draw.

Table 4

Estimated systolic blood pressure (SBP) in mmHg at age 3 years in children of mothers with and without anemia or low iron intake in the 1st or 2nd trimester of pregnancy

	Mean (SE) SBP		Mean (SE) SBP	Difference (mmHg)	95% Confidence Interval	P-value
Anemia	No	Yes				
1st Trimester	97.5 (3.4)	99.1 (3.0)	-1.6	-5.0, 1.8	0.36	
2nd Trimester	97.3 (3.4)	97.0 (3.6)	0.3	-1.7, 2.3	0.76	
Low iron intake	No	Yes				
1st Trimester	97.4 (3.0)	96.8 (3.0)	0.5	-0.6, 1.7	0.36	
2nd Trimester	95.1 (3.1)	95.7 (3.1)	-0.6	-2.3, 1.1	0.50	

Anaemia in pregnancy is defined according to criteria published by the Centers for Disease Control and Prevention²⁶ as haemoglobin level <11 g/dl in the first trimester and <10.5 g/dl in the second trimester. Low iron intake is defined as <27 mg/day, the Recommended Dietary Allowance during pregnancy.²⁵ Estimates are adjusted for blood pressure measurement conditions (extremity used, cuff size, state, position, measurement order), child gestational age, sex and age, height and weight at the time of blood pressure measurement; and for maternal age, pre-pregnancy body mass index, pregnancy weight gain, smoking, income, education and race/ethnicity. SE is standard error.